

Transforming and Combining Functions

MATH 145 – Applied Calculus

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More Complex Functions

Last time, we discussed some of the basic types of functions.

We also discussed compositions, one of the ways to combine simpler functions to make more complicated ones.

There are more ways to combine or change functions.

Piecewise Functions

Piecewise Function

A piecewise function is a function that has different formulas on different intervals.

Example: The absolute value function $f(x) = |x|$ returns x if x is positive, and $-x$ if x is negative. We can write this as:

$$f(x) = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Piecewise Functions

Example:

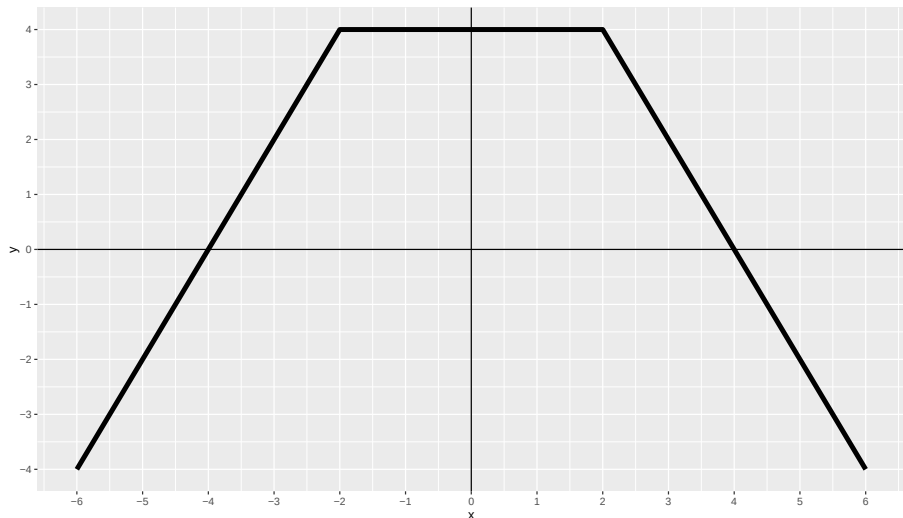
$$f(x) = \begin{cases} x + 3 & \text{if } x < 2 \\ x^2 & \text{if } 2 \leq x \leq 4 \\ 2x - 4 & \text{if } x > 4 \end{cases}$$

Evaluate $f(1)$, $f(2)$, $f(3)$, $f(4)$ and $f(5)$.

Graph $f(x)$.

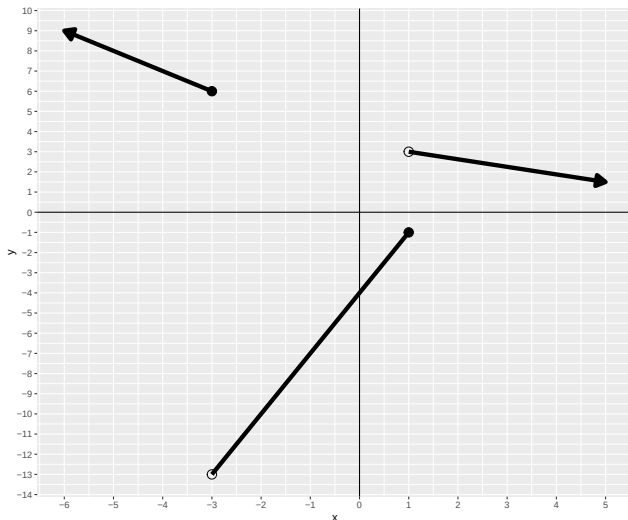
Piecewise Functions

Example: Determine the formula for the piecewise function shown below:



Piecewise Functions

Example: Determine the formula for the piecewise function shown below:



Combining Functions with Algebraic Operations

You can perform algebraic operations on functions, just like you can perform algebraic operations on numbers.

Algebraic Operations for Functions

Let f and g be functions that share the same domain. Then,

1. The **sum of f and g** is the function $f + g$ defined by
$$(f + g)(x) = f(x) + g(x).$$
2. The **difference of f and g** is the function $f - g$ defined by
$$(f - g)(x) = f(x) - g(x).$$
3. The **product of f and g** is the function $f \cdot g$ defined by
$$(f \cdot g)(x) = f(x) \cdot g(x).$$
4. The **quotient of f and g** is the function $\frac{f}{g}$ defined by
$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$
 for all x such that $g(x) \neq 0$.

Combining Functions with Algebraic Operations

This is mainly about notation, but this notation will allow us to simplify functions for calculus formulas. If you can notice when a function is a combination of two simpler ones, it may help make future things easier.

Algebraic Operations for Functions

Let f and g be functions that share the same domain. Then,

1. The **sum of f and g** is the function $f + g$ defined by
$$(f + g)(x) = f(x) + g(x).$$
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4. The **quotient of f and g** is the function $\frac{f}{g}$ defined by
$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \text{ for all } x \text{ such that } g(x) \neq 0.$$

Piecewise Functions

Example: Let $f(x) = 2x + 3$, $g(x) = x^3$ and $h(x) = \sin(x)$. (Assume all functions share the domain of $\mathbb{R}$.)

Find:

1. $(f + g)(x)$.
2. $(h - f)(x)$.
3. $(g \cdot f)(2)$.
4. $\left(\frac{f}{g}\right)(5)$. Where is this undefined?
5. $\left(\frac{f}{h}\right)(x)$. Where is this undefined?
6. $(h \cdot (g - f))(x)$.

Translating Functions

Suppose we have the function $f(x) = x^2$. If I want to move the whole graph 3 units upwards (i.e. in the positive y -direction), how could I do it?

We want to send (x, y) to $(x, y + 3)$.

Translating Functions

Vertical Translation of a Function

Given a function $y = f(x)$ and a real number a , the transformed function $y = g(x) = f(x) + a$ is a **vertical translation** on the graph of f . That is, every point $(x, f(x))$ on the graph of f gets shifted vertically to the corresponding point $(x, f(x) + a)$ on the graph of h .

Translating Functions

Suppose we have the function $f(x) = x^2$. Now if I want to move the whole graph 3 units to the right (i.e. in the positive x -direction), how could I do it?

We want to send (x, y) to $(x + 3, y)$.

Translating Functions

Horizontal Translation of a Function

Given a function $y = f(x)$ and a real number b , the transformed function $y = h(x) = f(x + b)$ is a **horizontal translation** on the graph of f . That is, every point $(x, f(x))$ on the graph of f gets shifted horizontally to the corresponding point $(x + b, f(x))$ on the graph of h .

Translating Functions Example

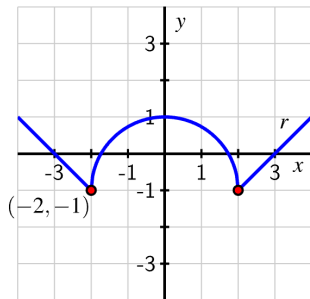


Figure 1.8.11. A parent function r .

Example: On the same axes as the plot of r , sketch the following graphs: $y = f(x) = r(x) + 2$, $y = g(x) = r(x + 1)$ and $y = h(x) = r(x + 1) + 2$. Be sure to label the point on each of f , g and h that corresponds to $(-2, -1)$ on the original graph of r .

Translating Functions Example

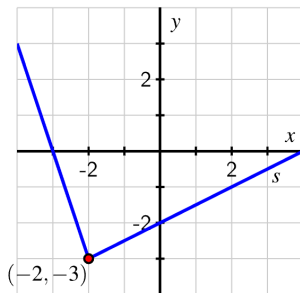


Figure 1.8.12. A parent function s .

Example: On the same axes as the plot of s , sketch the following graphs: $y = j(x) = s(x) - 1$, $y = k(x) = s(x - 2)$ and $y = \ell(x) = s(x - 2) - 1$. Be sure to label the point on each of j , k and ℓ that corresponds to $(-2, -3)$ on the original graph of s .

Scaling Functions

Vertical Scaling of a Function

Given a function $y = f(x)$ and a real number $c > 0$, the transformed function $y = v(x) = cf(x)$ is a **vertical stretch** of the graph of f . Every point $(x, f(x))$ on the graph of f gets stretch vertically to the corresponding point $(x, cf(x))$ on the graph of v . If $0 < c < 1$, the graph of v is a compression of f toward the x -axis; if $c > 1$, the graph of v is a stretch of f away from the x -axis. Points where $f(x) = 0$ are unchanged by the transformation.

Given a function $y = f(x)$ and a real number $c < 0$, the transformed function $y = v(x) = cf(x)$ is a reflection of the graph of f across the x -axis followed by a vertical stretch by a factor of $|c|$.

Scaling Functions Example

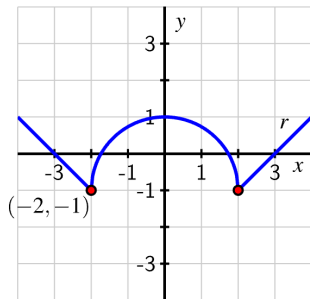


Figure 1.8.11. A parent function r .

Example: On the same axes as the plot of r , sketch the following graphs: $y = f(x) = 3r(x)$ and $y = g(x) = \frac{1}{3}r(x)$. Be sure to label the point on each of f and g that corresponds to $(-2, -1)$ on the original graph of r .

Scaling Functions Example

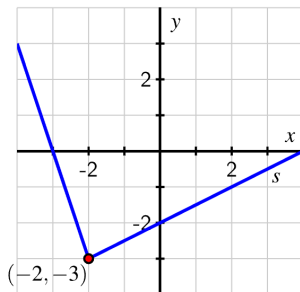


Figure 1.8.12. A parent function s .

Example: On the same axes as the plot of s , sketch the following graphs: $y = j(x) = -2s(x)$ and $y = k(x) = -\frac{1}{2}s(x)$. Be sure to label the point on each of j and k that corresponds to $(-2, -3)$ on the original graph of s .

Scaling Functions

Horizontal Scaling of a Function

Given a function $y = f(x)$ and a real number $c > 0$, the transformed function $y = h(x) = f(cx)$ is a **horizontal stretch** of the graph of f . Every point $(x, f(x))$ on the graph of f gets stretch vertically to the corresponding point $(x, f(cx))$ on the graph of h . If $0 < c < 1$, the graph of v is a compression of f toward the y -axis; if $c > 1$, the graph of h is a stretch of f away from the y -axis. Points where $x = 0$ are unchanged by the transformation.

Given a function $y = f(x)$ and a real number $c < 0$, the transformed function $y = h(x) = f(cx)$ is a reflection of the graph of f across the y -axis followed by a horizontal stretch by a factor of $|c|$.

Scaling Functions Example

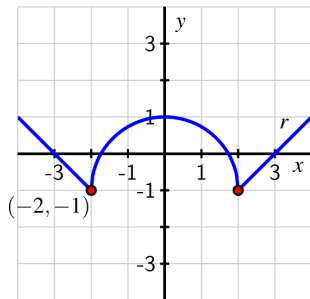


Figure 1.8.11. A parent function r .

Example: On the same axes as the plot of r , sketch the following graphs: $y = f(x) = r(3x)$ and $y = g(x) = r(\frac{1}{3}x)$. Be sure to label the point on each of f and g that corresponds to $(-2, -1)$ on the original graph of r .

Scaling Functions Example

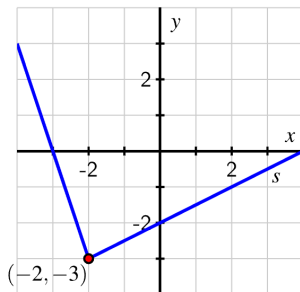
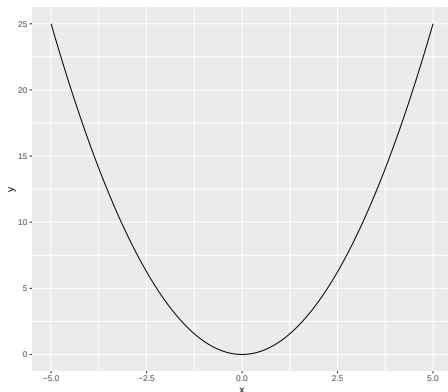


Figure 1.8.12. A parent function s .

Example: On the same axes as the plot of s , sketch the following graphs: $y = j(x) = s(-2x)$ and $y = k(x) = s(-\frac{1}{2}x)$. Be sure to label the point on each of j and k that corresponds to $(-2, -3)$ on the original graph of s .

Translating Functions Example



Example: Above is the graph of $y = f(x) = x^2$. What does the transformation $y = g(x) = 2f\left(\frac{1}{2}x + 3\right) - 4$ do to the graph of f ?